

MRI Foreground/Background Detection

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Abstract

This note outlines how we derive foreground and background masks for MRI volumes in RadQy. Each section documents one detection method, its assumptions, and the clean-up steps needed so FG/BG pixels are reproducible across datasets.

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1 Introduction

This document lists the foreground/background (FG/BG) segmentation methods in `backend/seg/mri`. Each section mirrors a Python module and notes its inputs, thresholds, morphology, and fallbacks so downstream IQMs get consistent masks.

2 Image Processing Segmenters

Classical pipelines based on intensity statistics and morphology.

2.1 OtsuHull (`OtsuHull.py`)

Legacy variant mirroring the original hybrid Otsu pipeline with more defensive fallbacks [1]. This subsection is listed first because it matches the earliest implementation shipped with RadQy.

Algorithm 1: OtsuHull foreground/background estimation

1. Input slice I ; cast to uint16.
 2. Histogram-equalize I to get H .
 3. Otsu on I and H to obtain masks m_1, m_2 and FG fractions w_1, w_2 .
 4. Hybrid recombination: $S = w_1 \cdot I + w_2 \cdot H$ (normalize to image range).
 5. Otsu on S to get final mask M .
 6. Cleanup: convex hull of M (fallback to hole filling if hull fails).
 7. Outputs: $fg = M, bg = 1 - M$.
- **Preprocess:** Cast to uint16. Histogram-equalize (uses `scikit-image` if available; otherwise internal equalization on $[0, 1]$).
 - **Dual thresholds:** Otsu on raw and equalized images to get masks and FG fractions w_1, w_2 .
 - **Re-threshold:** Weighted sum of raw and equalized images, followed by Otsu.
 - **Hull cleanup:** Convex hull if available; otherwise hole-filling. Any exception yields FG all-ones, BG all-zeros.
 - **Output:** `segment(img)` returns FG; `make_masks(img)` returns (fg, bg) in $\{0, 1\}$.

3 Deep Learning Segmenters

Learned models for FG/BG inference.

3.1 U-Net (placeholder)

Planned slot for the learned U-Net FG/BG mask generator. Add architecture details (depth, channels, skips), preprocessing steps, and post-processing (e.g., thresholding, hole filling) when the model is integrated.

4 Notes for Integration

All segmenters operate per 2D slice and return uint8 masks in $\{0, 1\}$. Background is always defined as the complement of foreground to ensure downstream IQMs receive matching shapes and disjoint FG/BG sets.

References

- [1] Amir Reza Sadri et al. “MRQy: An open-source tool for the quality control of structural MRI images”. In: *Journal of Imaging* 6.1 (2020). Placeholder entry; replace with canonical citation if available, pp. 1–15.